



Agentic AI Foundation

AGNTCon + MCPCon China

Where the **agentic stack** is being built.



Coordination Engineering

The next leap in AI Agent engineering

From governing one agent to governing a team

Orchestration has not converged

Widely built; the control point is still contested.

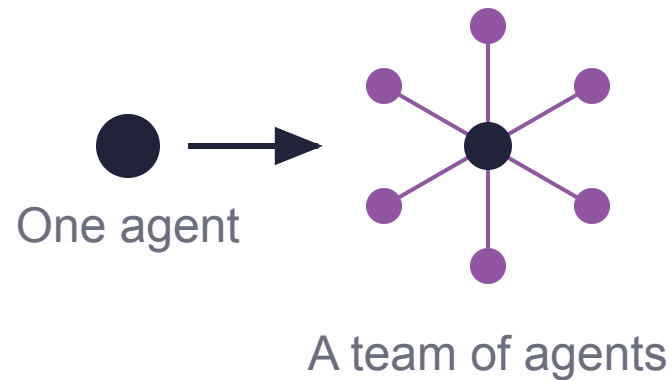
Multi-agent orchestration

- **Hierarchy and graph orchestration**
- **Chat room and handoff**
partial context visibility
- **Team collaboration**
organised by tasks and issues
- **Human in the centre**
from initiator to participant and governor

In practice: Claude Code Agent Teams · Multica · Paperclip · WorkBuddy

More agents, four amplified costs

Capability grows — and so do four costs.



Token and scheduling overhead



Context continuity and drift
information lost when summarising



Coordination failure
and task dependency



Verification and ownership
who accepts the result

Enterprise settings add distributed state, security isolation and semantically consistent recovery.

The window is not the usable space

Model, harness and task decide the effective space.

1D Lossy management	Compression · Rollback · Fine-grained control
2D Lateral expansion	Spawn · Cross · Aggregate Cross exchanges facts between parallel contexts mid-run
3D Branching	Fork
4D Time	Evolve · Memory

$$\text{effective space} = \min(\text{ECL}_{90}, \text{window} - \text{reserved})$$

reserved = system prompt + tool/MCP definitions + message history + output reserve
200K–500K is a priority verification range, not an industry-wide conclusion

Coordination Engineering

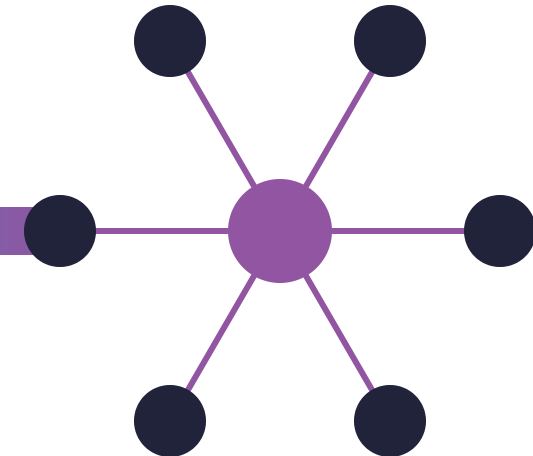
Harness Engineering extended, not replaced.

Harness Engineering



+ collaboration

Coordination



Coordination adds

Planning · Division of labour · Communication
Verification · Recovery · Reusable experience

openJiuwen's route

WorkSwarm AgentTeam is the runtime base.



Autonomous collaboration

preemptive scheduling
emergent teamwork



SwarmFlow

coded control flow
deterministic orchestration



SwarmChat

with Team Organization
shared decision space



SwarmSkill

with Swarm Memory
reusable experience

WorkSwarm AgentTeam

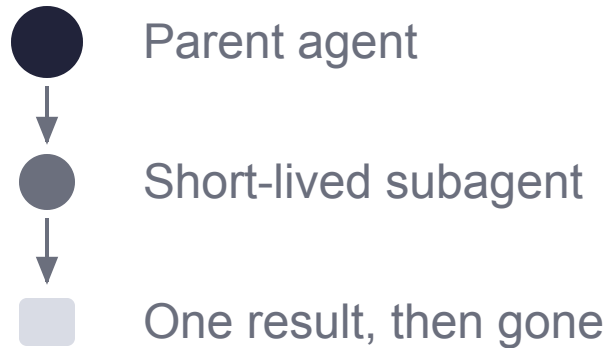
multi-agent runtime collaboration

Human participation is shown as HITS — Human in the Swarm.

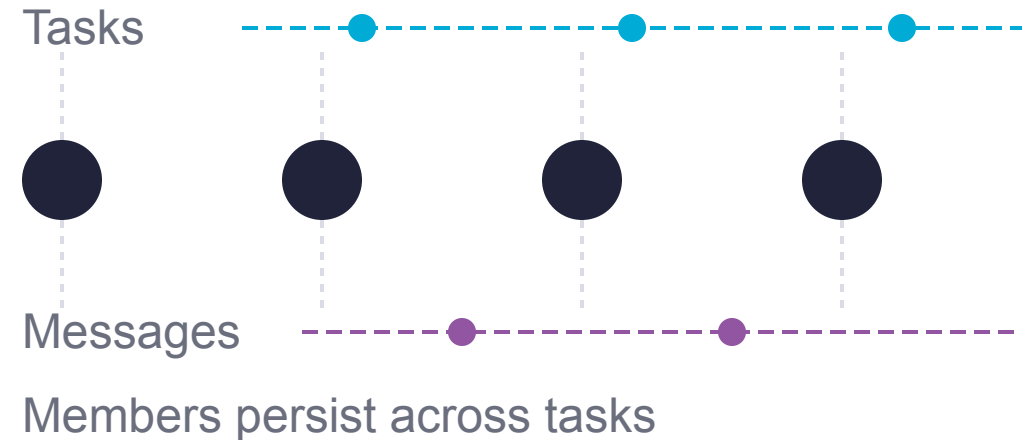
The subagent becomes a teammate

Context stops being an input and becomes a state.

One-shot subagent



Long-lived teammate



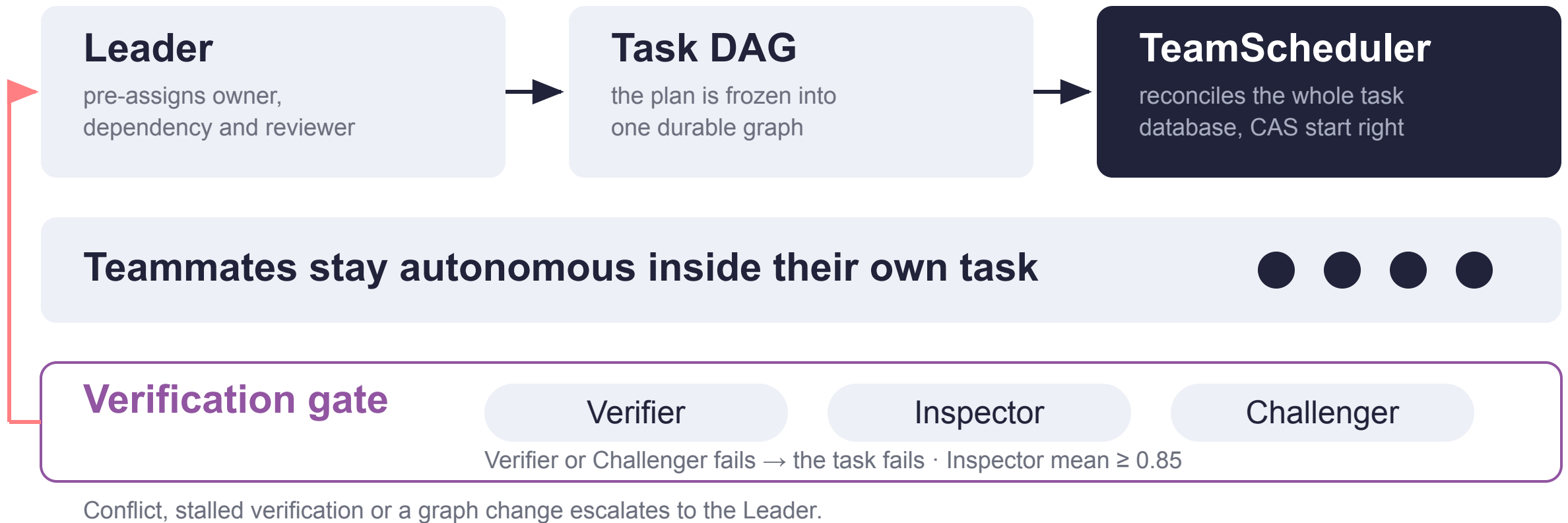
Worktree isolation
reversible parallel coding



Team Workspace
shared artefacts

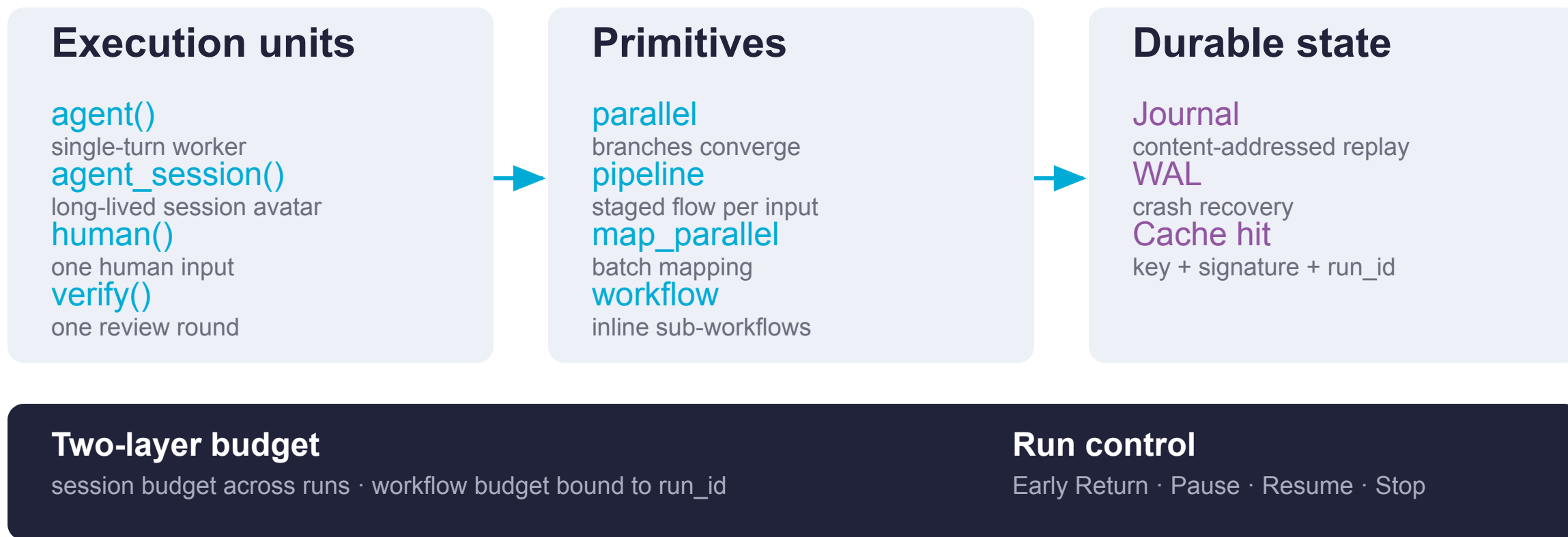
The plan becomes a task graph

The Leader leaves the steady-state hot path.



Code carries the control flow

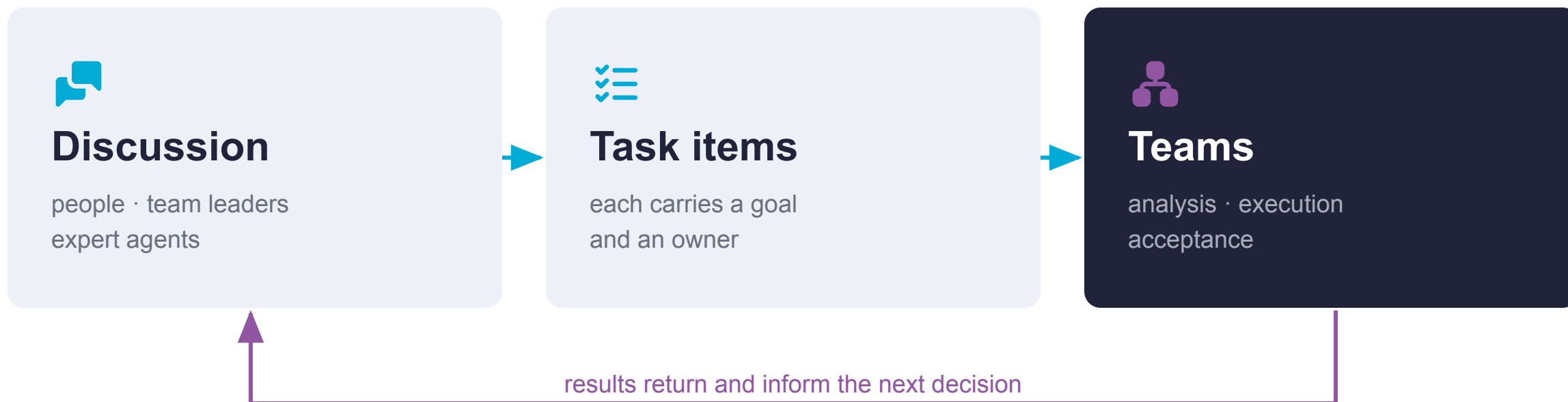
Deterministic orchestration out of the agent loop.



Stop seals the current run and cannot be resumed.

Discussion becomes tasks

A shared space wired straight into delivery.



SwarmChat is a shared decision and task-generation entry, not a cross-team background task protocol.

Composing harnesses, not calls

A CLI agent harness is a whole runtime, not a model.

WorkSwarm

CLI

MCP

Skill



Claude Code

own agent loop
own context
own tools and skills



Codex

own agent loop
own context
own tools and skills



DeepSeek Harness

own agent loop
own context
own tools and skills



Expert RSI

bad cases → prompt section · skill · tool · rail, inside one expert harness

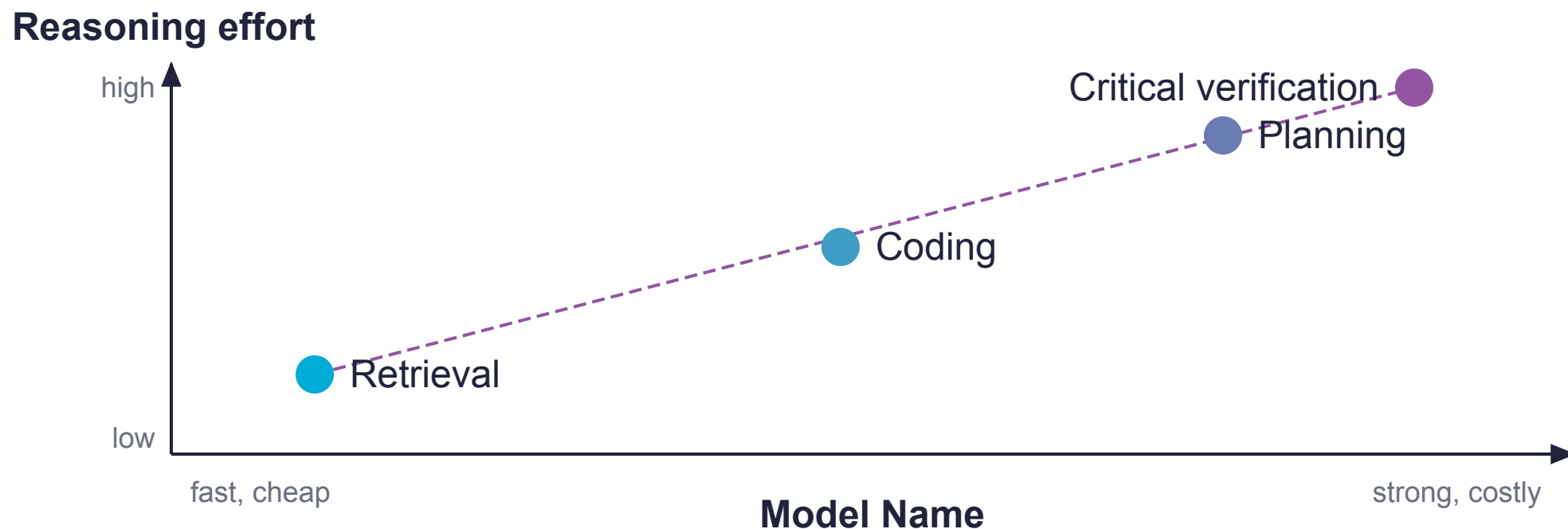
DeepSeek Harness is abbreviated DSH. Expert RSI does not change team structure.

Where the **agentic stack** is being built.

Model and reasoning effort

Choose both per task and per member.

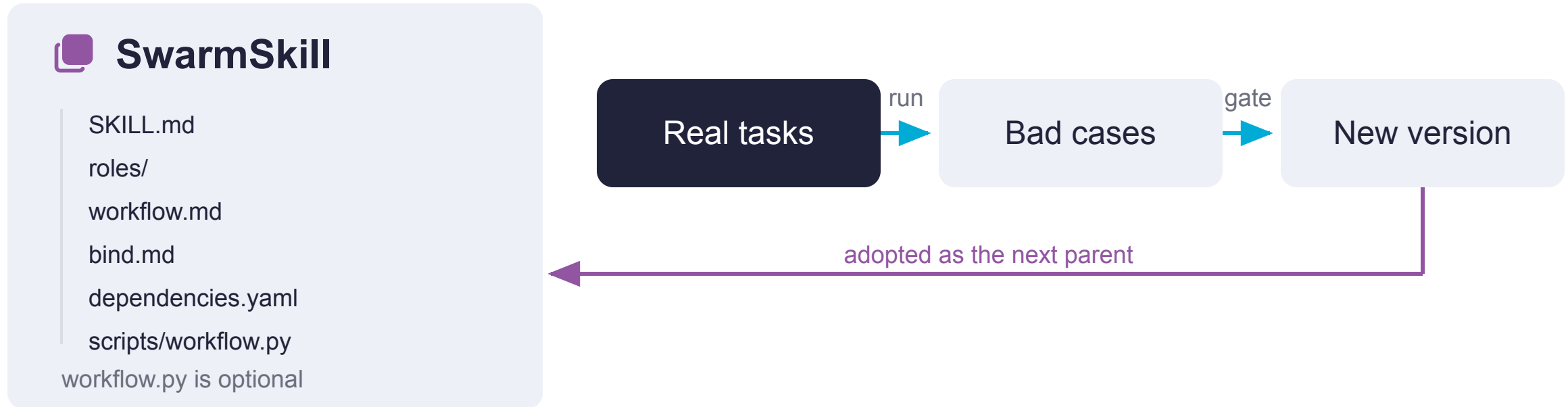
🔧 Task difficulty + member role → one routing decision



Model selection is in the Team Model Pool; reasoning effort travels with the model request config.
Dynamic two-axis scheduling is a direction, not a fully landed capability.

Collaboration becomes a package

Proven collaboration, captured and versioned.



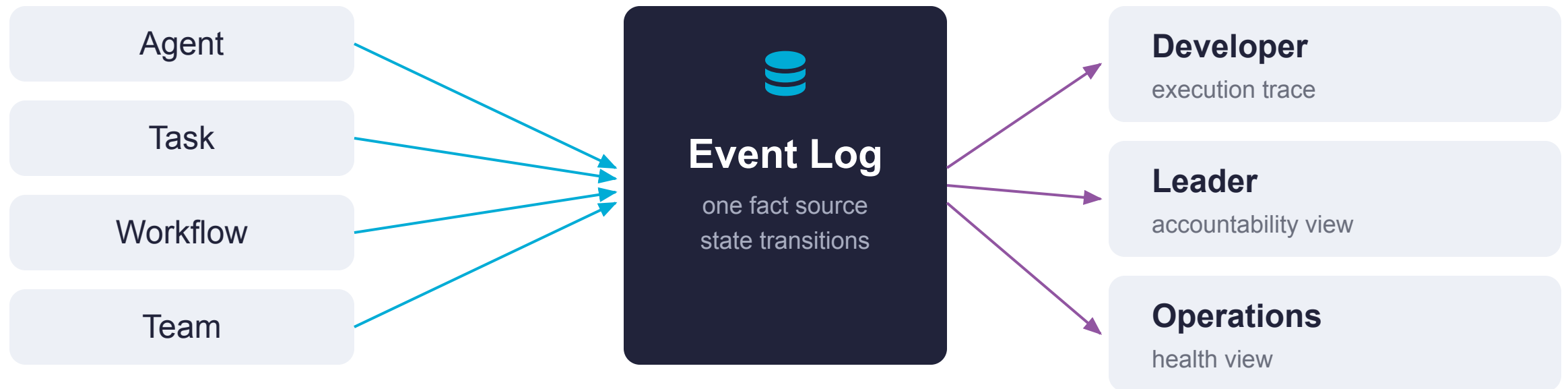
Attributable text gradients locate what to adjust: prompt, context, memory, tool or skill.

Shared and discovered through Hub or Symphony.

An open-collaboration SwarmSkill may ship without workflow.py.

One event log, three readings

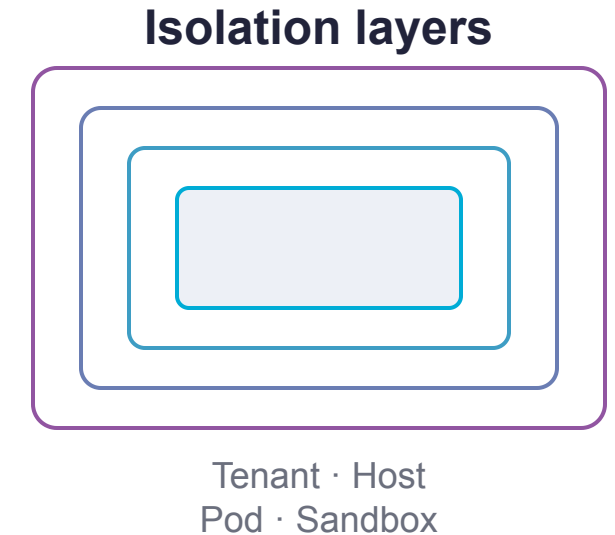
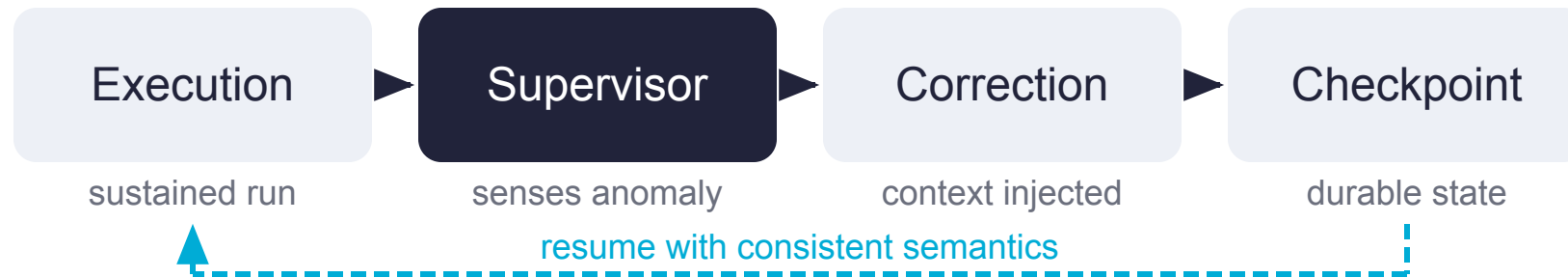
One fact source, projected per role.



The business accountability tree and the technical trace are linked but do not substitute.
One task may retry across several traces.

Supervise, correct, recover

Long tasks survive interruption, not by luck.

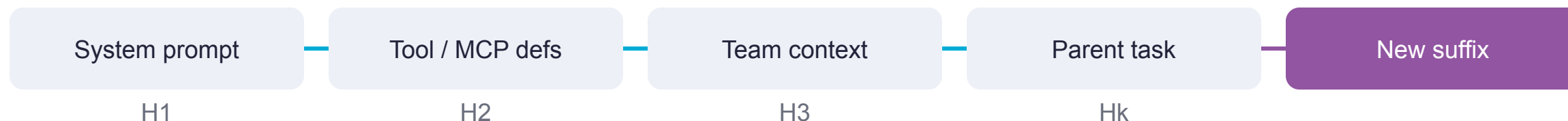


Tiered scheduling · task parallelism · automatic elasticity

Million-scale session concurrency, thousand-scale swarm instances and millisecond wake are distributed-runtime platform figures, not one AgentTeam's performance.

Reuse the prefix, follow the cache

Reuse the stable prefix; send only what is new.



$$H_0 = \text{Hash}(\text{model}, \text{tokenizer}, \text{template version}), H_i = \text{Hash}(H_{i-1}, \text{Block}_i)$$

Full request

Block 1 ... Block k + new suffix

On a prefix hit

prefix_hash = Hk + new suffix

22%+

lower first-token latency (verified)

20%+

higher prefix cache hit (verified)

Architecture level: heterogeneous scheduling cost -50%, throughput +10%.

Chained hash is an experimental direction, not yet implemented, and is excluded from these results.

openJiuwen architecture panorama



Agent Applications

WorkSwarm · iCode · Science Discovery · Symbiosis · Others

Hyper Agent WorkSwarm

SwarmSkill · SwarmFlow · Swarm Memory · UI · TUI

Agent Framework

Agent Gateway

Unified entry · IAM · RBAC · Channels

Agent Core

RSI Engine · Swarm Engine · Harness Engine

Agent Memory

Refinement · Hybrid retrieval · Evolution

Agent Distributed Runtime

Agent Serving · Distributed Executor · Agent Protocol

Agent System Service

Sandbox · Memory Store · Agent UI
Infer · Online Learning · Model Routing

Agentic Hubs

Agent Hub

Skill Hub

MCP Hub

Plugin Hub

Agent Studio

Zero-code & low-code

Application Space

Workflow Canvas

Skills & Plugins

Resource Management

Models

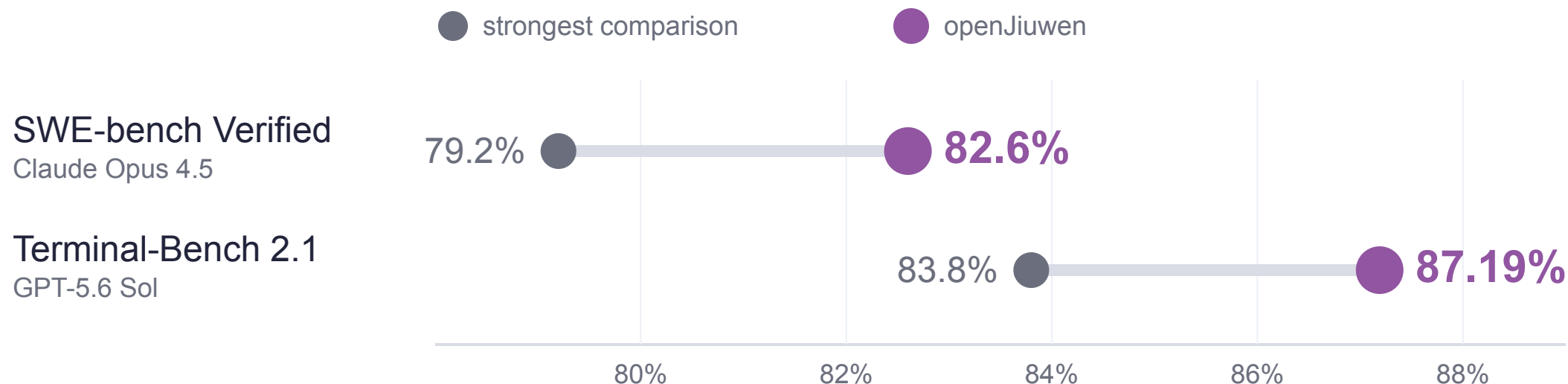
Infrastructure

Hardware · Chipsets

Where the **agentic stack** is being built.

The harness moves the result

Same model — the architecture still moves the result.



Structural Composability

inner loop: observe · reason · act · verify
outer loop: goal · plan · execute · update
rails: safety · memory · planning · tools

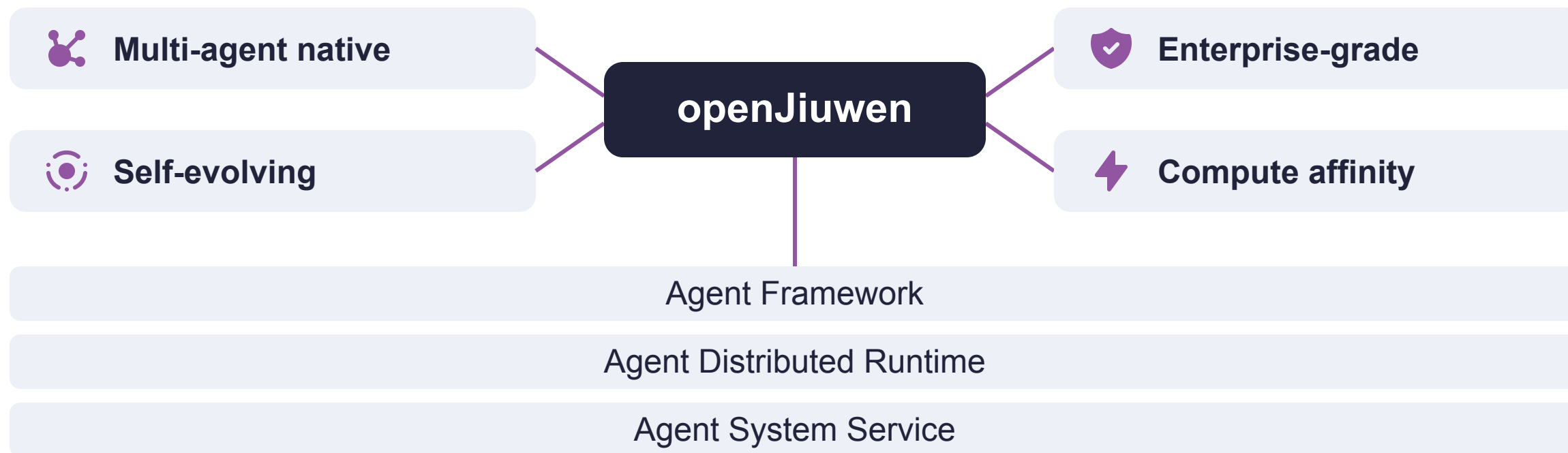
Runtime Adaptivity

goal mode: completion · blockage · budget
LSP diagnostics injected actively
context management · self-reflection

Data time point 2026-09-01. Paper: Beyond Static Harnesses for Long-Horizon Coding Agents.
Shipped in WorkSwarm on HarmonyOS, Windows and macOS.

A general-purpose Agent OS

Four properties, one platform.



In production

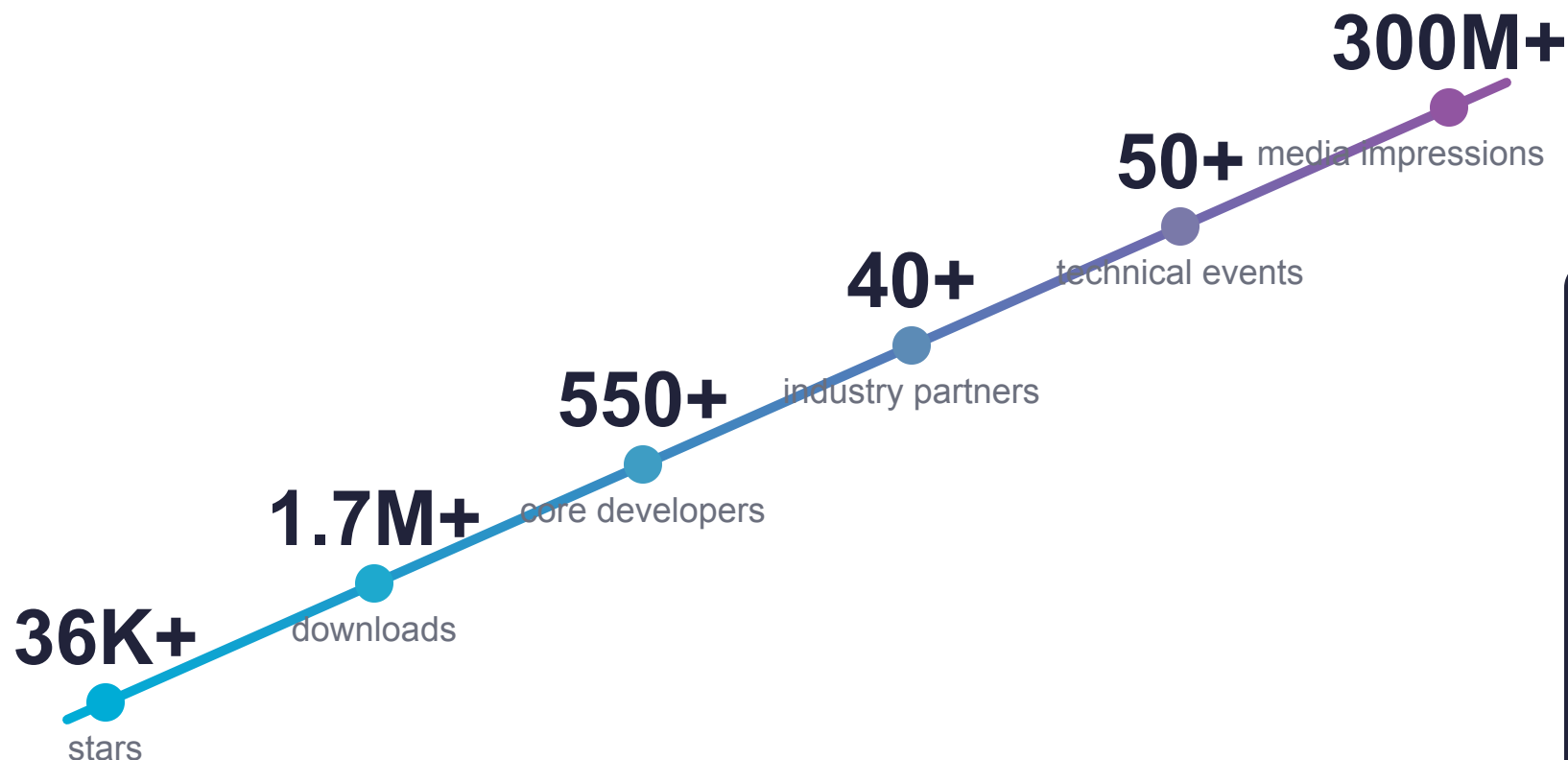
Postal Savings Bank of China · USTC science intelligence · HarmonyOS Xiaoyi · Huawei Cloud OfficeAce

10+ industries and 50+ commercial customer deployments (DataFun caliber).

Where the **agentic stack** is being built.

Website and community

One snapshot, one place to start.



Start here

openjiuwen.com

atomgit.com/openJiuwen

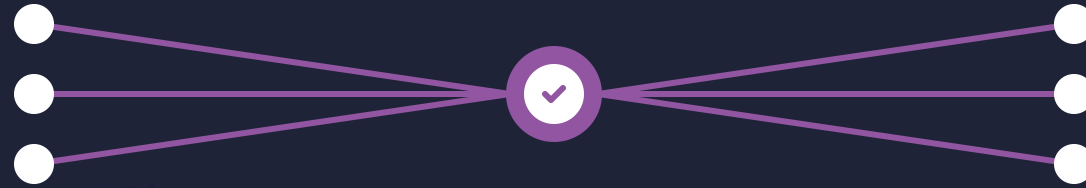
github.com/openJiuwen-ai



Linux Foundation AAIF
gold member

Community figures use the DataFun 7-24 caliber and are not mixed with earlier material.

Where the **agentic stack** is being built.



THANK YOU

WorkSwarm

openjiuwen.com

Coordination completes when the team converges